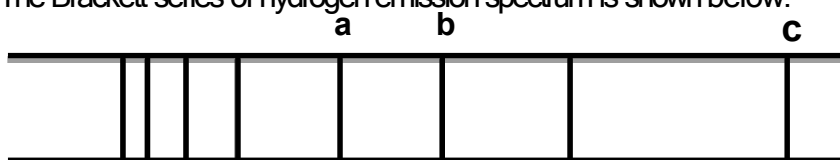


HOUR : 1

1.
 - a) Describe Bohr's atomic postulates. (CLO1, C1)
 - b) Explain the difference between ground state and excited state. (CLO1, C2)
2. Explain why the emission spectrum of hydrogen atom is a line spectrum. (CLO1, C2)
3. Give the difference between line spectrum and continuous spectrum. (CLO1, C2)
4.
 - a) The first line of Balmer series has a wavelength of 656.3 nm. Calculate its frequency. (CLO3, C4)
($4.57 \times 10^{14} \text{ s}^{-1}$)
 - b) Calculate the wavelength (in nm) and frequency of the fifth line in Balmer series. (CLO3, C4)
($\lambda = 397.04 \text{ nm}$, $\nu = 7.56 \times 10^{14} \text{ s}^{-1}$)
 - c) Calculate the energy emitted by a hydrogen atom during the formation of the second line in Lyman series. (CLO3, C4)
(Energy emitted = $1.94 \times 10^{-18} \text{ J}$)
5. The Brackett series of hydrogen emission spectrum is shown below.



- a) In what region does this line exist? (CLO1, C1)
- b) Sketch the electronic transition of lines **a**, **b** and **c** on the energy level diagram of the hydrogen atom. (CLO3, C3)
- c) Which of the lines **a**, **b** and **c** has the shortest wavelength? Calculate the wavelength of the radiation that produce the line. (CLO3, C4)
($\lambda = 1.95 \times 10^6 \text{ m}$)
- d) Explain why successive line in Lyman series become closer to the previous line until the line form a continuum. (CLO1, C2)

HOUR : 2

6. Calculate the longest wavelength for an electron transition in the Paschen series. (CLO3, C4)
($\lambda = 1.88 \times 10^6 \text{ m}$)
7. An excited electron in the $n = 7$ level of hydrogen atom drops to a lower energy level, emitting light of 397 nm.
 - a) Calculate the value of n for the level to which the electron drops? (CLO3, C4)
 - b) Determine the energy of the electron (CLO3, C3)
 - i) before it drops to the lower energy level
 - ii) when it resides at the lower energy level
 - c) Calculate the energy emitted for the transition? (CLO3, C4)
(Energy emitted = $5.01 \times 10^{-19} \text{ J}$)

8. **FIGURE 1** shows the Balmer series of hydrogen emission spectrum.

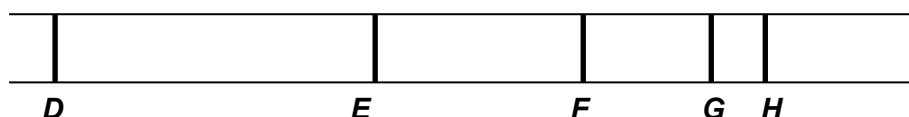


FIGURE 1

- Determine the electron transition that produces line **E**.
- Explain the formation of line **E**.
- Calculate the energy of photon that produces line **F** with the wavelength of 434 nm.

MID SEM 2011 / 2012 (CLO3, C4)

(Energy emitted = 4.58×10^{-19} J)

9. In the hydrogen atom, an electron transition from a higher to a lower energy level emits a photon with a wavelength of 1282 nm in Paschen series. Determine the energy level before it drops to produce this transition and calculate its energy.

MID SEM 2013 / 2014 (CLO3, C3)

(-8.72×10^{-20} J)

10. The excited electrons of a group of irradiated hydrogen atoms are randomly scattered to $n = 1, 2, 3$ and 4 energy levels. Sketch an energy level diagram to show the transitions for all possible lines produced when all these electrons drop to lower levels. Identify the electron transition which gives the shortest wavelength.

PSPM 2008 / 2009 (CLO3, C4)

11. Calculate the first **three** wavelengths of the possible transitions of an electron in the Paschen series for a hydrogen atom. Show these wavelengths in a sketch of the line spectrum for this emission series.

(CLO3, C3)

PSPM 2015 / 2016 (CLO3, C4)

($\lambda = 1875$ nm, $\lambda = 1282$ nm, $\lambda = 1094$ nm)

HOUR : 3

12. **FIGURE 2** shows the Lyman series of the hydrogen line spectrum.

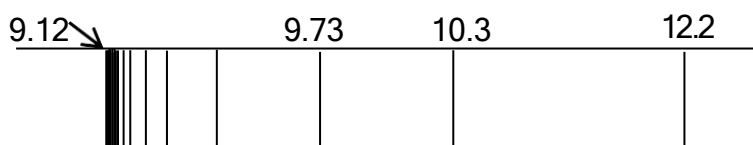


FIGURE 2

Wavelength ($\times 10^{-8}$ m)

Calculate the ionisation energy (kJ mol^{-1}) of the hydrogen atom by using **FIGURE 2**.

List two species that exhibit similar line spectrum as that of hydrogen atom.

(CLO3, C4)

(1312 kJ mol^{-1})

13. A hydrogen atom requires a minimum energy of $2.18 \times 10^{-18} \text{ J atom}^{-1}$ to remove an electron from its ground state level. Determine whether a blue violet light with a wavelength of 434.0 nm can affect this process.

PSPM 2008 / 2009 (CLO3, C3)

14. Calculate the ionisation energy of hydrogen atom in kJ mol^{-1} . **MID SEM 2014 / 2015 (CLO3, C4)**
(1312 kJ mol^{-1})

15. a) Give two weaknesses and advantages of Bohr's atomic model. (CLO1, C2)
b) State de Broglie's postulate on the dual nature of electron and Heisenberg's uncertainty principle. (CLO1, C1)

HOUR : 4

16. a) Define orbital. (CLO1, C1)
b) Name the four quantum numbers of an electron in an orbital and state the characters determined by each quantum numbers. (CLO1, C1)
c) For each set of the following quantum numbers,

set	P	q	r
n	2	3	5
l	0	2	1

- i) Identify the type of orbitals. (CLO3, C4)
ii) Determine the number of orbitals for p , q and r ? (CLO3, C4)
iii) Determine the maximum number of electrons that can occupy the subshell? (CLO3, C4)
iv) Write one set of possible quantum numbers for an electron in orbital r . (CLO3, C4)
v) Sketch and label the 3-D shapes for all orbitals q . (CLO3, C3)
17. Which of the following combination of n and l representing real orbitals and which are impossible?
a) $2d$
b) $7s$
c) $3f$
d) $4p$ (CLO3, C4)
18. Which of the following (n, l, m, s) quantum numbers are not allowed? Explain your answer.
a) $(1, 1, 0, +\frac{1}{2})$
b) $(3, 1, -2, +\frac{1}{2})$
c) $(2, 1, 0, +\frac{1}{2})$
d) $(2, 0, 0, +\frac{1}{2})$ (CLO3, C4)
19. Write the electronic configuration and draw orbital diagram for each of the following atoms/ions.
a) Al
b) Al^{3+}
c) Cl
d) Cl^- (CLO3, C3)
20. Determine the number of electrons in a zinc atom with
a) orbital quantum number, l , equals to 1.
b) magnetic quantum number, m , equals to -1 .
c) $l = 3$ and $m = +1$
d) $s = -\frac{1}{2}$ **PSPM 2009 / 2010 (CLO3, C4)**

HOUR : 5

21. Chromium is an element in d block of the periodic table. Write the electronic configuration of chromium. Explain the anomalous electronic configuration in chromium. Write a set of quantum numbers for an electron in 3d orbital. **MID SEM 2005 / 2006 (CLO3, C3)**
22. Copper is the ninth element in the first row of d-block in the Periodic table.
 a) Write the electron configuration of copper according to the Aufbau principle and the actual configuration as determined by experiment.
 b) Explain the anomalous electronic configuration of Copper. **PSPM 2004 / 2005 (CLO3, C4)**
23. Fe is a transition element, which can form Fe^{2+} and Fe^{3+} ions in aqueous solution.
 a) Draw orbital diagram of Fe.
 b) Write the set of quantum number for an electron in the 4s orbital of Fe.
 c) Write the electronic configuration of Fe^{2+} and Fe^{3+} ions. Explain which of these two ions is likely to be more stable. **(CLO3, C3)**
24. Two orbital diagrams for the electronic configuration shown below are not allowed. State the principles that are not obeyed and explain.
- a)

↑	↑	↑	↑
2s	2p		
- b)

↑↑	↑	↑	↑
2s	2p		

MID SEM 2014 / 2015 (CLO3, C4)
25. The higher energy electrons in atom **Q** have the principal quantum number, $n = 3$ and the angular momentum quantum number, $l = 1$. Atom **Q** will form Q^{3+} ion in order to achieve octet.
 a) List the sets of quantum numbers for each electron in outermost shell.
 b) State the number of electrons in **Q** and Q^{3+} .
 c) Write the electronic configuration and the orbital diagram for **Q**. State the number of orbital that are filled with unpaired electron.
 d) Give one difference between electron with $n = 4, l = 0$ and $n = 4, l = 1$. **(CLO3, C4)**
26. $n = 3, l = 1, m = -1, s = +\frac{1}{2}$
 $n = 3, l = 1, m = -1, s = -\frac{1}{2}$
 $n = 3, l = 1, m = 0, s = +\frac{1}{2}$
 $n = 3, l = 1, m = +1, s = +\frac{1}{2}$

The given set of quantum numbers represents the four outermost electrons of an unknown element **R** at ground state.

- a) Give the proton number of the element.
 b) Draw a complete orbital diagram for **R**. How many unpaired electrons would be found?
 c) Suggest an ion that may be formed from this element. Write its electronic configuration.
 d) Name the quantum number that indicates the shape of the orbital. Sketch the shape of the orbitals of the four electrons above. **(CLO3, C3)**